

Heaven's Light is Our Guide

RAJSHAHI UNIVERSITY OF ENGINEERING & TECHNOLOGY
RAJSHAHI-6204, BANGLADESH



A CASE STUDY REPORT ON
Task Allocation and Optimal Labor Cost in Garment Manufacturing: A
Case Study

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DEPARTMENT OF INDUSTRIAL & PRODUCTION ENGINEERING
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ON

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Case Study**

This is to certify that the case study entitled “**Task Allocation and Optimal Labor Cost in Garment Manufacturing: A Case Study**” is carried out by Md. Saidul Islam Mridul, Roll No.: 2105028 and S M Ashfaque Hossain, Roll No.: 2105009, under my supervision in the Department of Industrial & Production Engineering of Rajshahi University of Engineering & Technology.

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CANDIDATES' DECLARATION

It is hereby declared that this case study or any part of it has not been submitted elsewhere for the award of any degree or diploma.

Md. Saidul Islam Mridul

S M Ashfaque Hossain

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Authors

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ABSTRACT

The garment manufacturing sector in Bangladesh with the majority of small and medium enterprises that work under strict cost margins and customer deadlines generally sets the task distribution and unit price depending on the experience and approximate calculations but not on the actual level of operator skills and wage differentials. In this research, a Binary Integer Linear Programming (BILP) model of the optimal allocation of the tasks and minimization of labor costs in garment SMEs which consider the heterogeneity of operators skills, wage rates, processing time needs, availability of equipment, and workload balance constraints are proposed. A real-life case study at the Noman Group, one of the largest apparel manufacturers in Bangladesh, with 34 operators and 19 assembly components in a T-shirt assembly line was used to validate the model. To determine the impact of the capacity utilization parameter on workforce engagement and cost, two scenarios were tested with a different capacity utilization parameter. The findings indicate that the BILP model not only decreases labor cost by about 15-15.6 percent of the current practice in the manual allocation but also provides the convenience to either restrict the number of operators utilized to the minimum possible cost or utilize the entire workforce at an insignificant difference in cost. The observations prove that the inclusion of skill level and wage data in planning decisions results in quantifiable benefits in the use of resources and cost-effectiveness. The conclusion is that the model, despite its simplicity in formulation, is highly versatile and effective in addressing real-world problems related to task allocation and optimization of costs when it comes to garment manufacturing. The model can also be further generalized to include stochastic demand, multi-product lines and other social goals and can be instantiated as an easy to use application to planners who need not know programming or operations research.

Keywords: apparel manufacturing; integer linear programming; task allocation; cost optimization; integer formulation

Table of Contents

ACKNOWLEDGEMENT.....	iv
ABSTRACT	v
Table of Contents	vi
List of Tables.....	vii
List of Figures	viii
CHAPTER 1: INTRODUCTION	9-11
1.1 Introduction	9
1.2 Problem Statement	10
1.3 Objectives.....	11
CHAPTER 2: LITERATURE REVIEW	12-15
2.1 Review of Literature.....	12
CHAPTER 3: METHODOLOGY	16-22
3.1 Mathematical Model	16
CHAPTER 4: RESULT & DISCUSSION	23
4.1 Data Collection.....	23
4.2 Result analysis.....	26
4.3 Discussion	30
CHAPTER 5: CONCLUSION & FUTURE WORK	32-33
5.1 Conclusion.....	32
5.2 Future Work	32
REFERENCES	34

List of Tables

Table no.	Title	page no.
Table 1:	Operational data	23
Table 2:	Detailed process data	24
Table3:	Data of operators	24

List of Figures

Table no.	Title	page no.
Figure 1:	Optimal task allocation (scenario 1)	26
Figure 2(a):	Income of operators (scenario 1)	27
Figure 2(b):	Capacity of operators (scenario 1)	27
Figure 3:	Optimal task allocation (scenario 2)	28
Figure 4(a):	Income of operators (scenario 2)	29
Figure 4(b):	Capacity of operators (scenario 2)	29

CHAPTER 1

INTRODUCTION

1.1 Introduction

The apparel sector of Bangladesh has got a greater attention than any other industry for its rapid growth and major share in foreign exchange earnings. Bangladesh is the second largest apparel exporter in the world after China [4]. The whole sector depends heavily on people doing the work, especially sewing and assembling which are very labor-intensive, since automation is not that advanced yet [6]. Small and medium enterprises handle most of the apparel chain. They do sub-contracting, Cut-Make-Trim services, and short production runs for brands and wholesalers. Everything is under pressure from costs and tight deadlines [5][15]. Labor costs are one of the biggest parts, so how workers are assigned, their skills, and the machines used really affects in making profit or not [5].

In the CMT model, buyers tell them the delivery dates, styles, and prices they want, FOB style. Then the SMEs have to organize cutting, sewing, finishing with what they have, and margins are super thin [3][6]. Using capacity efficiently is given much importance by apparel companies to operate cost-effectively and provide timely service to their clients. Delivering orders on time is essential to improve relationship with customers. Reducing operational costs and delivering orders on time are closely linked to improving line efficiency[9]. But planning prices and production often just uses what customers say about time, how complex the design is, and rough guesses from experience and not really looking at their own labor and limits formally [2]. This leads to problems, like promising too much capacity, wasting time on tasks, and then rushing with overtime or outsourcing. That jacks up costs and hurts profit [2]. Studies from Bangladesh, Turkey show using math programming for production can boost profit 20 to 60 percent, and cut costs over 30 percent [4][12]. Which shows how much waste is hidden in the usual ways.

A central issue for these SMEs is that managers and supervisors do not have training in operations research or optimization tools [2]. They use rules of thumb, basic stopwatch timing, or just what they know from before, not real models [8]. Even projects on balancing lines or improving productivity stick to simple heuristics or simulations, not exact math tools.

So differences in skills, how people learn, wages, those are not modeled much. But quality and output depend on how good the tailors and sewers are [2][5]. Recent study points out that labor skills matter more than machines in small firms. Yet models treat workers the same or just aim for max efficiency, not min cost, with different skills and pay [2].

The objective of the Case Study is to address the gap between optimization models and real-world planning practices in garment SMEs. This study implements binary integer programming model to allocate tasks to operators of different skills and with different pay scales within limited processing times and balance constraints to achieve maximum reduction in labor cost within the given time for delivery. The case study illustrates, using practical factory data, that a simple integer programming model can be incorporated within an application software even without programming or mathematical knowledge to obtain minimum labor cost and optimal task allocation.

1.2 Problem Statement

Garment making in Bangladesh usually means dealing with huge volumes of work. It also involves quick turnaround times. Plus there is intense rivalry when it comes to keeping prices low and operations running smoothly. It is pretty clear that getting the production lines balanced right plays a big role. It helps hit those productivity goals without wasting too many resources. Still, most factories in the sector face ongoing issues. Things like uneven workloads across teams stand out. Some workstations sit idle more than they should. And workers or machines end up overworked just to meet the daily output needs.

Noman Group, one of Bangladesh's largest apparel manufacturer, deals with several manufacturing lines which requires careful allocation of resource and demand satisfaction. Within their existing operating condition, in a t-shirt assembly line- operator and machinery task distribution depends mainly on manual approaches, repeatedly causing operational inefficiencies and imbalance throughout the assembly process. These complexity not only increase manufacturing cost as well as labor cost but also causes postponements and downtime of various operations to address these inefficiencies.

To face these operational inefficiencies, this research proposes implementation of a Binary Integer Linear Programming (BILP) model to optimize garment production line at Noman Group. The suggested model takes essential operational limitations into account while reducing the overall cost of the production.

1.3 Objectives

The aim of our study is to achieve the following objectives.

- a. To observe the current scenario of any production line performance and find major inefficiencies related to task and resource allocation.
- b. To develop a Binary Integer Linear Programming (BILP) model to improve the production line while respecting the available resource and reducing overall labor cost.

CHAPTER 2

LITERATURE REVIEW

2.1 Review of Literature

Line balancing and assembly line optimization have always been of interest, since the sewing line is a labor-intensive process and demarcates the heart of garment production. Mixed integer linear programming (MILP) and integer programming (IP) models have been suggested to both balance the workloads among workstations as well as increase throughput or minimize the amount of stations or idle time [1][2]. MILP has been applied to Garment Line Balancing where it is shown, by integrating resource capacity constraints along with lean methodology principles that there can be more than 50% reduction in labor cost, while having all the demands met and using all of the machinery at their maximum capacity limits [1][10]. Another IP model to balance a shirt line within man-power and machine limitation recorded higher effectiveness but with less number of work-stations than classical ranked positional weight heuristic, pushing mathematical programming to perform better on heuristics-based line balancing in garment manufacturing sector [9]. Complementing these models, simulation-based studies have tested alternative balanced-line configurations using discrete-event simulation to reveal the benefits in terms of output, line efficiency, and labor productivity, they also quantify reductions in labor cost and increases in revenue when line balancing is combined with simulation [7].

In addition to the micro-level line balancing stream, there is research contributing to the macro level of production planning and product-mix optimization in textile and apparel industry. Linear programming optimization models have been formulated to jointly minimize labor, inventory and subcontracting costs over medium-term planning horizons considering sector specific features like fabric shrinkage, waste, new workers' efficiency and training time [3]. The LP-based scheduling models have been found to increase profits (by up to 22–60%) and reduces costs (by over one-third) by adjusting the distribution of processing times, materials and labor among cutting, sewing and finishing tasks [4][11]. These works highlight how quantitative decision-making takes heed of scarce resources and can promote efficient competitive advantage in low-cost apparel-exporting countries. Nevertheless they do aggregate

labor (total available hours per department) and not operators with different skills and wage rates.

There is a smaller, although quite significant literature directly on the issue of minimizing labor costs in textile and clothing. Dynamic programming has been applied to the planned size and costs of labor force in contract sewing factories, where labor is the biggest cost element and profit depends not only on volume but also on quality and delivery-on-time [5]. Dynamic programming problem can be used for the optimization of the number of workers and labor costs over planning horizons. However, it requires expertise in Mathematical models which are not easily applicable in SMEs. Simulation modeling for garment sewing lines have also focused on optimizing labor cost by rearranging work division, changing operators' distribution and maximizing machine's load where the simulation was applied to compare alternative allocation of labor [8].

More related with the focus of task allocation with heterogeneous workers, in workforce-planning studies humans aspects such as skills, training, personalities and motivation have been incorporated into multi-objective optimization models [15]. *Othman et al.* introduced a nonlinear programming model that minimizes hiring, firing, training, and overtime costs as well as the dismissing rate of highly productive workers, this study created an interactive decision support system by using an Excel–LINGO interface to reach end users [15]. This study shows how heterogeneous workers can be incorporated in optimization models, and turned into decision-aids. For the particular problem of small and medium size garment manufacturing, *V. Nguyen et al* provided general binary integer formulations for task assignment and pricing using labor skill and wage levels in planning to achieve processing-time constraints and balanced workloads on operators [2]. The model was verified through case studies and was applied in a stand-alone MATLAB-based desktop application that has been developed for use by non-programmers, which illustrated that the integer programming can be both relevant and efficient in addressing the allocation problems faced by apparel SMEs even under relatively simple modelling settings.

In the general manufacturing literature, integer linear programming variables are often used in scheduling (e.g. job-shop problems) and resource allocation, but computational difficulty is a common bottleneck. Recent research on ILP formulations of job-shop scheduling has shown that reformulated models with a much smaller number of variables and tighter constraints can

dramatically reduce solution times and provide near-optimal solutions within tight computational bounds, often aided by decomposition and coordination techniques [16]. In the manufacture of apparel and garments, integer linear programming and mixed-integer linear programming (utilized in conjunction with algorithms like branch-and-bound) have been applied to production planning and capacity-allocation problems to show that exact methods are better able to provide more optimal production plans when compared manual or current company practice [1][17].

Meanwhile, metaheuristic algorithms- Genetic Algorithm, Simulated annealing and Particle Swan Optimization have been extensively investigated in production systems for complex optimization purposes such as matrix production system, and collaborative disassembly line [12]. Comparative studies indicate that these methods can obtain potentially enhanced performance in terms of metrics, and they usually outperform each other with respect to objective functions and problem nature. For example, it has been found that Simulated Annealing can attain better solutions with less computation time than Genetic Algorithm in matrix production optimization [18]. Hybrid algorithms were employed to superior control of multi-man robot collaborative disassembly-line balancing, based on NP-hard problems than classical ones such as GA and Jaya algorithm pairwise combinations [13]. However, these metaheuristics often have high requirements for tuning of parameters, dedicated software and expert knowledge burdens to limit their practical utility under limited resource garment planners in SMEs who may not possess computational competencies. Another relevant literature stream is related to lean manufacturing and modernization in apparel. Lean-derived methods focus on bottleneck detection, line balancing, and work-in-process/lead-time reduction by using VSM or CRP [11]. Push factors such as problems with internal production flows and process control have also been found in prior studies within both garment factories and apparel SMEs, which is reflected by the adoption of lean practices, an approach known to improve line efficiency and reduce cycle times leading to enhanced performance although smaller firms face impediments due poor managerial support, shortages of resources such as low capital outlay, training costs and a lack of knowledge [6][10]. The integration of lean methodology with linear programming, including A-MILP to lean line-balancing frameworks, has been demonstrated to improve efficiency and cost-effectiveness (though again much of the focus has been on maximizing

system throughput and minimizing waste rather than modeling detail of skill- and wage-driven labor cost [1].

Two major constraints have been found from these literatures. Firstly, many optimization models consider the workers to be from a single pool of identical or homogeneous workers or treat labor as just total available hours without consideration for inter- individual differences in skill, learning and wage which may exist especially in small garment shops where experienced tailors and helpers work each at different pay scales. When skills are taken into account, they are generally handled as qualitative criteria of assignment rather than as explicit budget and constraint functions. Secondly, even when advanced optimization or metaheuristic models have been created, these models often do not move beyond academic prototypes that is to say they do not move towards user interactive tools that SME planners and managers can use without appropriate professional instruction [2]. Even where they do for Excel-based or LINGO-linked interfaces, the topic is likely more involved in aggregate workforce planning and product-mix optimization than detail level task allocation within garment sewing lines [15].

In this sense, the current case study is at the crossroad of the optimization of garment production, reduction of labor cost and practical decision support for SMEs. It extends binary integer formulations that have been developed in the areas of task assignment within small apparel companies to include real practices of garment making, it is concerned with minimizing labor costs and includes both skill level as well as wage level as key planning parameters. The study attempts to show that mathematically sound structurally simple models can be made available to practitioners by employing real production data from a garment SME in form of process-time across operations, operator skill levels, wage bands, and order sizes. Solving the binary/integer programming model to balance the workloads while adhering to task time constraints. From the literatures, this approach addresses explicitly these limitations: avoids the heavy computational and tuning efforts required by advanced metaheuristics, relaxes homogeneous labor assumptions, as well as granting a tool to be used by planners who are not familiar with programming or operations research. In the process, this case empirically illustrates that optimization-based task assignment is also a viable means for driving down costs and achieving leaner economic production.

CHAPTER 3

METHODOLOGY

Binary Integer Programming (BIP) is a type of mathematical optimization where decision variables can take only two values: 0 or 1. It is used to model yes/no, on/off, or select/reject decisions in complex problems such as routing, scheduling, and resource allocation. By restricting variables to binary values, BIP allows precise representation of logical constraints and combinatorial choices, making it a powerful tool in operations research and decision making.

3.1 Mathematical Model

Decision variables:

SM_{ij}	binary variable indicating the capability of operator i to perform component j .
X_{ij}	integer variable indicating the quantity of component j that shall be produced by operator i
Y_{ij}	binary variable indicating whether operator i is assigned to component j

Indices:

i	index of operators $I \in \{1, 2, 3, \dots, n\}$
j	index of components $j \in \{1, 2, 3, \dots, m\}$

Parameters:

SL_i	skill level of operator i $SL_i \in \{1, 2, 3, 4, 5\}$
DL_j	difficulty level of component j $DL_j \in \{1, 2, 3, 4, 5\}$
W_i	hourly wage of each operator i (TK)
C_j	cycle time of a component j (seconds)
O_j	number of semiproducts in each process (pcs)
D	duration given by customer (days)

OPT	order processing time (days)
APT	total available production time (days)
E_j	quantity of available equipment for component j
W_t	working time per day (hours)
Cap	target capacity level for all operators (%)
B_j	Minimum batch size required for component j
T_i^{max}	Maximum number of components that can be assigned to operator i
P_{jk}	precedence indicator; $P_{jk}=1$. if component j must precede component k (j is upstream of k), otherwise 0

There are some assumptions:

- A process can be handled by one or more operators
- There is initially no work in the progress buffer
- There is no parallel process.

Objective function: The objective is to allocate tasks to operators for optimal labor cost, and the objective function can be presented as:

$$\text{Minimize, } Z = \sum_{i=1}^n \sum_{j=1}^m X_{ij} * \frac{C_j * W_j}{3600}$$

Expanding the sum:

This sums over all operators i and components j. For example, if we have n=3 operators and m=4 components, the expanded form is:

$$\begin{aligned} Z = & X_{11} * \frac{C_1 * W_1}{3600} + X_{12} * \frac{C_2 * W_1}{3600} + X_{13} * \frac{C_3 * W_1}{3600} + X_{14} * \frac{C_4 * W_1}{3600} \\ & + X_{21} * \frac{C_1 * W_2}{3600} + X_{22} * \frac{C_2 * W_2}{3600} + X_{23} * \frac{C_3 * W_2}{3600} + X_{24} * \frac{C_4 * W_2}{3600} \\ & + X_{31} * \frac{C_1 * W_3}{3600} + X_{32} * \frac{C_2 * W_3}{3600} + X_{33} * \frac{C_3 * W_3}{3600} + X_{34} * \frac{C_4 * W_3}{3600} \end{aligned}$$

Constraints:

1. Demand satisfaction

This constraint enforces that, for every component j , the total quantity produced by all capable operators equals the required order quantity for that component. Operators with $SM_{ij} = 0$ (not capable) contribute nothing, so only capable operators can satisfy the demand of each process.

$$\sum_{i=1}^n X_{ij} * SM_{ij} = O_j, \quad ; \forall j \in J$$

Expanding the sum:

For $n=3$ operators and $m=4$ components, the expanded form is:

Expanded form for component $j=1$,

$$X_{11}SM_{11} + X_{21}SM_{21} + X_{31}SM_{31} = O_1$$

Expanded form for component $j=2$,

$$X_{12}SM_{12} + X_{22}SM_{22} + X_{32}SM_{32} = O_2$$

Expanded form for component $j=3$,

$$X_{13}SM_{13} + X_{23}SM_{23} + X_{33}SM_{33} = O_3$$

Expanded form for component $j=4$,

$$X_{14}SM_{14} + X_{24}SM_{24} + X_{34}SM_{34} = O_4$$

2. Operator time limit

This ensures that the total working time assigned to each operator i does not exceed that operator's available production-time fund (total available days times working hours per day). It links assigned quantities, cycle times, and the calendar/time capacity of each worker.

$$\sum_{j=1}^m X_{ij} * SM_{ij} * \frac{C_j}{3600} \leq APT * W_t \quad ; \forall i \in I$$

where $APT = D - OPT$ Minimum

Expanding the sum:

For $m=4$ components and $n=3$ operators, The total working time for operator $i=1$ would be:

$$X_{11} * SM_{11} * \frac{C_1}{3600} + X_{12} * SM_{12} * \frac{C_2}{3600} + X_{13} * SM_{13} * \frac{C_3}{3600} + X_{14} * SM_{14} * \frac{C_4}{3600} \leq APT * W_t$$

Similarly for operator i=2 would be:

$$X_{21} * SM_{21} * \frac{C_1}{3600} + X_{22} * SM_{22} * \frac{C_2}{3600} + X_{23} * SM_{23} * \frac{C_3}{3600} + X_{24} * SM_{24} * \frac{C_4}{3600} \leq APT * W_t$$

Similarly for operator i=3 would be:

$$X_{31} * SM_{31} * \frac{C_1}{3600} + X_{32} * SM_{32} * \frac{C_2}{3600} + X_{33} * SM_{33} * \frac{C_3}{3600} + X_{34} * SM_{34} * \frac{C_4}{3600} \leq APT * W_t$$

3. Capacity / workload

This constraint forces every selected operator to be loaded at least up to a target utilization level *Cap* of their time capacity. It is used to avoid very low-load assignments and to promote workload balancing among the operators.

$$Cap \leq \sum_{j=1}^m X_{ij} * \frac{SM_{ij} * C_t}{3600 * APT * W_t} ; \forall i \in I$$

Expanding the sum:

For m=4 components and n=3 operators,

The capacity constraints for i=1 becomes,

$$Cap \leq \frac{X_{11} * SM_{11} * \frac{C_1}{3600} + X_{12} * SM_{12} * \frac{C_2}{3600} + X_{13} * SM_{13} * \frac{C_3}{3600} + X_{14} * SM_{14} * \frac{C_4}{3600}}{APT * W_t}$$

The capacity constraints for i=2,

$$Cap \leq \frac{X_{21} * SM_{21} * \frac{C_1}{3600} + X_{22} * SM_{22} * \frac{C_2}{3600} + X_{23} * SM_{23} * \frac{C_3}{3600} + X_{24} * SM_{24} * \frac{C_4}{3600}}{APT * W_t}$$

The capacity constraints for i=3,

$$Cap \leq \frac{X_{31} * SM_{31} * \frac{C_1}{3600} + X_{32} * SM_{32} * \frac{C_2}{3600} + X_{33} * SM_{33} * \frac{C_3}{3600} + X_{34} * SM_{34} * \frac{C_4}{3600}}{APT * W_t}$$

4. Machine/equipment limit

The number of operators involved in a component should be equal to or less than the amount of equipment available for that component. If there are, for example, two machines for a sewing

operation, at most two operators can be assigned to that operation. A binary decision variable Y_{ij} is needed so that if an operator i is assigned to a component j , Y_{ij} equals 1 and 0 otherwise.

$$\sum_{i=1}^n Y_{ij} \leq E_j \quad ; \quad \forall j \in J$$

Expanding the sum:

For $m=4$ components and $n=3$ operators,

For component $j=1$,

$$Y_{11} + Y_{21} + Y_{31} \leq E_1$$

For component $j=2$,

$$Y_{12} + Y_{22} + Y_{32} \leq E_2$$

For component $j=3$,

$$Y_{13} + Y_{23} + Y_{33} \leq E_3$$

For component $j=4$,

$$Y_{14} + Y_{24} + Y_{34} \leq E_4$$

5. Linking assignment and quantity

These “big-M” and lower-bound constraints connect the integer quantity X_{ij} with the binary assignment Y_{ij} . If an operator is not assigned to a component ($Y_{ij} = 0$), then X_{ij} must be 0; if $X_{ij} > 0$, then Y_{ij} is forced to 1, meaning the operator is formally assigned to that component

$$\begin{aligned} X_{ij} &\leq M * Y_{ij} & ; \quad \forall j \in J \text{ and } \forall i \in I \\ X_{ij} &\geq Y_{ij} & ; \quad \forall j \in J \text{ and } \forall i \in I \end{aligned}$$

6. Variable domain constraints

These specify that production quantities must be non-negative integers and assignment decisions must be binary. They ensure the model reflects discrete task allocations and operator assignments in the real sewing line

$$\begin{aligned} X_{ij} &\geq 0 \text{ and integer} \\ Y_{ij} &\in \{0, 1\} \end{aligned}$$

7. Minimum Batch Size per Task Assignment

micro-assignments increase setup time, reduce efficiency, and cause unnecessary task switching. To overcome this problem, a minimum batch size constraint is added. This constraint ensures that if a worker is assigned to a specific operation, the quantity produced must be at least a predefined minimum batch size.

$$X_{ij} \geq B_j * Y_{ij} \quad ; \forall i \in I$$

8. Maximum Tasks per Operator

An operator is capable of performing a certain number of tasks only. These are due to various real-life constraints such as training, cost involved, cognitive load, and specialization on sewing lines. This is because it would otherwise be difficult for an operator to handle many tasks simultaneously.

$$\sum_{j=1}^m Y_{ij} \leq T_i^{max} \quad ; \forall i \in I$$

9. Precedence

Guarantees conservation of flow in the chain of activities. The number of units produced by the downstream activity k cannot exceed those produced by the upstream activity j feeding into it. Thus, there will be no overproduction of subsequent activities while some are limiting, maintaining the logical order of production activities in the sewing process.

$$\sum_{i=1}^n X_{ik} \leq \sum_{i=1}^n X_{ij}$$

The problem is addressed through three main steps:

- First, the capability of each operator is determined;
- Next, the decision variables, objective function, and constraints are formulated
- Finally, the integer programming problem is solved.

The elements of the problem like the decision variables, objective functions are formulated separately a strategy that is often called problem-based approach. As the model would be implemented in MATLAB, all parameters are represented as vectors and matrices. x and y variables are equal to the number of operators and processes respectively. The lower and upper

limits of these variables also are represented as vectors and the value of these vectors is indicated in the preceding section. Objective functional and constraints are then formulated with regard to the mathematical model explained above and formulated into an optimization set-up. Lastly, the problem that has been obtained is solved with the help of the `intlinprog` function.

There can be more than one optimal solution to a linear integer problem. In this case, there can be multiple allocation plans (or vectors of x) with the same labor cost (or objective value). The difference among these solutions lies in the workload balance of operators. In this case algorithm strategy is to employ a constraint in such a way that the integer problem is required to be solved only once. In particular, the capacity level Cap added to Constraint (3) will make sure that all the operators will take part in the process at $Cap\%$ 0 to 100% of their capacity. Thus, the workload of operators will be carried out within a target range. This range will be greater with a smaller Cap , and the balance of work will not be the priority. Then, the most important will be the lowest labor cost and the number of participants, which is also met by the algorithm.

CHAPTER 4

RESULT & DISCUSSION

4.1 Data Collection

The Integer programming model has been applied for a real word case study in Bangladesh. Noman Group, which is one of the largest garment manufacturer in Bangladesh, was chosen to apply the model. The observation in this company shows that it has several garment product lines like T-shirt, polo-shirt and sweaters. A t-shirt manufacturing process includes 34 operators involved in a production with 19 assembly components. It is required to produce 10,000 product pieces, and the total available processing time target is 10 days. The production line works consecutively for 8h per day. And with the current setup the overall cost comes around 68,832 BDT. Also the minimum batch quantity is 500 meaning that if a worker is assigned to a specific operation, the quantity produced must be at least 500 units. To demonstrate the significance of the target capacity level(Cap), two scenarios with Cap at 0% and 0.1% will be presented.

Table 1: Operational data

Parameters	Inputs
Number of operators (n)	34 operators
Number of components (m)	19 assembly components
Demand(O)	10,000 pieces
Total available processing time (APT) days	10 days
Working time per day (Wt) hours	8 h
Target capacity level (Cap)	Scenario 1: 0 Scenario 2: 0.1

Difficulty level of each component, the labor wage, and the amount of available equipment are also added. These data are presented in Tables 2 and 3

Table 2: Detailed process data

j	Process Name	Difficulty level (DL)	Available equipment (E)	Cycle time (CT)
1	SHOULDER JOIN	4	13	18 s
2	RIB MAKE	3	13	9 s
3	RIB SERVICING	3	12	9 s
4	NECK JOIN	4	13	12 s
5	BACK TAPE JOIN	4	13	18 s
6	NECK TOP STITCH	3	13	9 s
7	LABEL POSITION MARK	2	7	9 s
8	LABEL TOP STITCH	3	9	18 s
9	BODY FOLD & SLEEVE MATCHING	2	14	9 s
10	SLEEVE JOIN	4	13	26 s
11	ARMHOLE TOP STITCH	3	9	23 s
12	THREAD CUT & BODY FOLD	1	14	9 s
13	SIDE JOIN	4	13	32 s
14	THREAD CUT & TURN	1	9	9 s
15	SLEEVE HEM	3	12	21 s
16	THREAD CUT	1	16	9 s
17	BODY HEM	3	12	30 s
18	THREAD CUT	1	16	9 s
19	LABEL MAKE	3	8	9 s

Table 3: Data of operators

Employee	skill level (SL)	hourly wage(W)
Operator 1	1	60
Operator 2	3	72
Operator 3	3	72
Operator 4	2	65
Operator 5	4	82
Operator 6	4	82
Operator 7	2	65
Operator 8	2	65
Operator 9	3	72
Operator 10	2	65
Operator 11	4	82
Operator 12	2	65
Operator 13	4	82
Operator 14	2	65
Operator 15	2	65
Operator 16	4	82

Operator 17	1	60
Operator 18	3	72
Operator 19	1	60
Operator 20	5	96
Operator 21	2	65
Operator 22	3	72
Operator 23	1	60
Operator 24	5	96
Operator 25	4	82
Operator 26	3	72
Operator 27	1	60
Operator 28	3	72
Operator 29	1	60
Operator 30	5	96
Operator 31	5	96
Operator 32	3	72
Operator 33	1	60
Operator 34	1	60

Operations	SHOULDER JOIN	RIB MAKE	RIB SERVICING	NECK JOIN	BACK TAPE JOIN	NECK TOP STITCH	LABEL POSITION MARK	LABEL TOP STITCH	BODY FOLD & SLEEVE MATCHING	SLEEVE JOIN	ARMHOLE TOP STITCH	THREAD CUT & BODY FOLD	SIDE JOIN	THREAD CUT & TURN	SLEEVE HEM	THREAD CUT	BODY HEM	THREAD CUT	LABEL MAKE
Operators																			
operator 1	-	10000	-	-	-	-	-	-	-	-	519	-	-	-	3203	-	-	-	10000
operator 2	-	-	-	-	-	-	-	10000	-	-	-	-	-	-	-	-	-	-	-
operator 3	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
operator 4	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
operator 5	-	-	-	-	1967	-	-	-	-	-	-	-	-	-	-	-	-	-	-
operator 6	-	-	-	-	-	-	-	-	-	9500	-	-	-	-	-	-	-	-	-
operator 7	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
operator 8	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
operator 9	-	-	-	-	-	-	-	-	-	-	-	-	-	-	500	-	7360	-	-
operator 10	-	-	-	-	-	-	-	-	10000	-	-	-	-	-	-	-	-	-	-
operator 11	-	-	-	10000	-	-	-	-	-	-	-	-	4350	-	-	-	-	-	-
operator 12	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
operator 13	6367	-	-	-	8033	-	-	-	-	-	-	-	-	-	-	-	-	-	-
operator 14	-	-	-	-	-	-	10000	-	-	-	-	-	-	-	-	-	-	-	-
operator 15	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
operator 16	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
operator 17	-	-	-	-	-	-	-	-	-	-	-	-	-	10000	-	-	-	-	-
operator 18	-	-	-	-	-	-	-	-	-	-	-	-	-	-	6297	-	-	-	-
operator 19	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
operator 20	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
operator 21	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
operator 22	-	-	-	-	-	-	-	-	-	-	9481	-	-	-	-	-	-	-	-
operator 23	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
operator 24	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
operator 25	3633	-	-	-	-	-	-	-	-	500	-	-	5650	-	-	-	-	-	-
operator 26	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
operator 27	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	10000	-	-	-
operator 28	-	-	10000	-	-	10000	-	-	-	-	-	-	-	-	-	-	2640	-	-
operator 29	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	10000	-
operator 30	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
operator 31	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
operator 32	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
operator 33	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
operator 34	-	-	-	-	-	-	-	-	-	-	-	10000	-	-	-	-	-	-	-

4.2 Result analysis

The task allocation solutions for the two scenarios are presented in figures 1 and 3. With no target capacity level, the model suggests involving 17 operators in the production while excluding almost other half of the operators depending on the capability of the worker and the difficulty of the task. Which makes the labor cost approximately 58,131 BDT and the income average and capacity average are BDT 3086 and 60% respectively.

Operations Operators	SHOULDER JOIN	RIB MAKE	RIB SERVICING	NECK JOIN	BACK TAPE JOIN	NECK TOP STITCH	LABEL POSITION MARK	LABEL TOP STITCH	BODY FOLD & SLEEVE MATCHING	SLEEVE JOIN	ARMHOLE TOP STITCH	THREAD CUT & BODY FOLD	SIDE JOIN	THREAD CUT & TURN	SLEEVE HEM	THREAD CUT	BODY HEM	THREAD CUT	LABEL MAKE
operator 1	-	10000	-	-	-	-	-	-	-	-	519	-	-	-	3203	-	-	-	10000
operator 2	-	-	-	-	-	-	-	10000	-	-	-	-	-	-	-	-	-	-	-
operator 3	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
operator 4	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
operator 5	-	-	-	-	1967	-	-	-	-	-	-	-	-	-	-	-	-	-	-
operator 6	-	-	-	-	-	-	-	-	-	9500	-	-	-	-	-	-	-	-	-
operator 7	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
operator 8	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
operator 9	-	-	-	-	-	-	-	-	-	-	-	-	-	-	500	-	7360	-	-
operator 10	-	-	-	-	-	-	-	-	10000	-	-	-	-	-	-	-	-	-	-
operator 11	-	-	-	10000	-	-	-	-	-	-	-	-	4350	-	-	-	-	-	-
operator 12	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
operator 13	6367	-	-	-	8033	-	-	-	-	-	-	-	-	-	-	-	-	-	-
operator 14	-	-	-	-	-	-	10000	-	-	-	-	-	-	-	-	-	-	-	-
operator 15	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
operator 16	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
operator 17	-	-	-	-	-	-	-	-	-	-	-	-	-	10000	-	-	-	-	-
operator 18	-	-	-	-	-	-	-	-	-	-	-	-	-	-	6297	-	-	-	-
operator 19	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
operator 20	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
operator 21	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
operator 22	-	-	-	-	-	-	-	-	-	-	9481	-	-	-	-	-	-	-	-
operator 23	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
operator 24	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
operator 25	3633	-	-	-	-	-	-	-	-	500	-	-	5650	-	-	-	-	-	-
operator 26	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
operator 27	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	10000	-	-	-
operator 28	-	-	10000	-	-	10000	-	-	-	-	-	-	-	-	-	-	2640	-	-
operator 29	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	10000	-
operator 30	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
operator 31	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
operator 32	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
operator 33	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
operator 34	-	-	-	-	-	-	-	-	-	-	-	10000	-	-	-	-	-	-	-

Figure 1: Optimal task allocation (scenario 1)

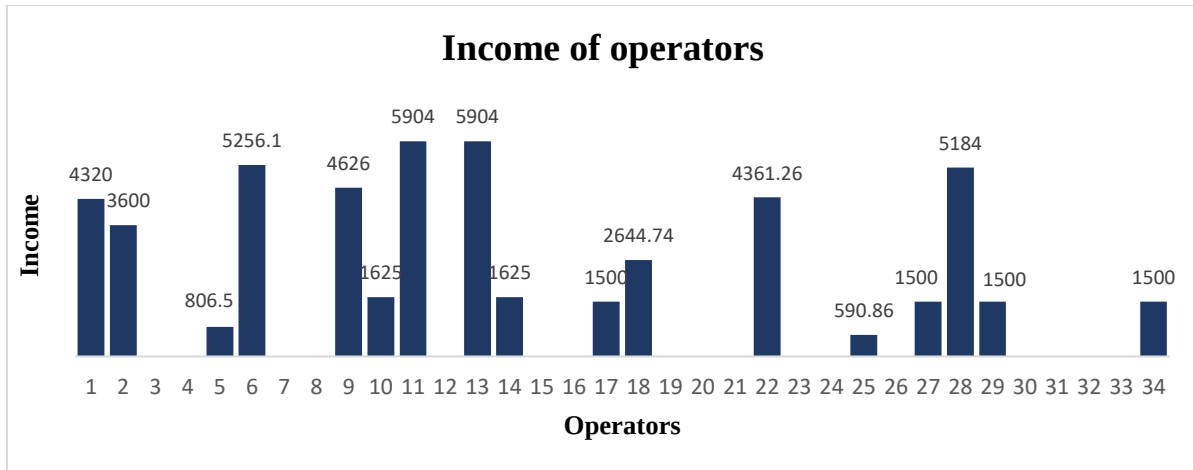


Figure 2(a): Income of operators (scenario 1)

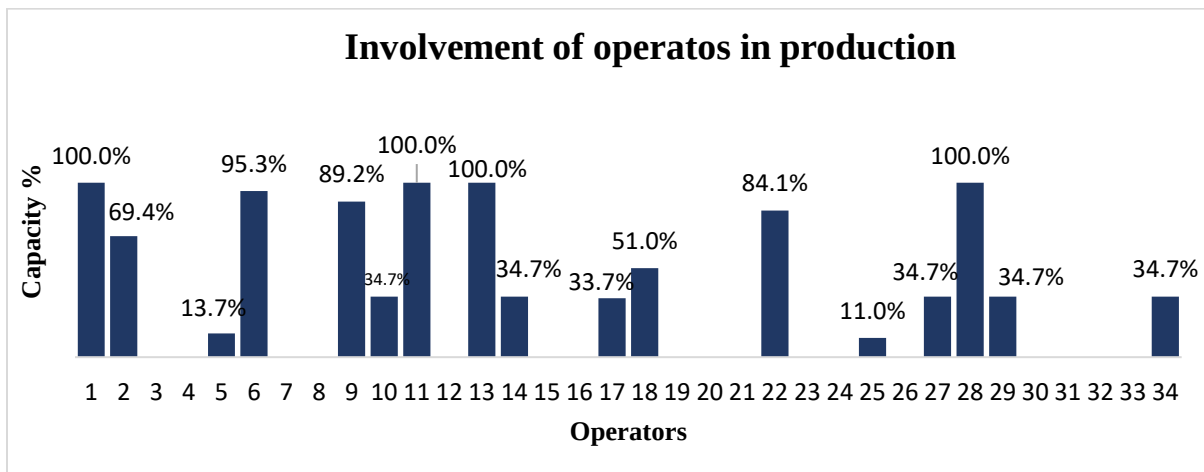


Figure 2(b): Capacity of operators (scenario 1)

The financial outcome of this allocation, as detailed in Figure 2(a), shows that there's a significant difference in individual earnings among the active operators. Where operator 11 earns the highest and operator 25 earns the lowest. The substantial difference in income distribution shows that operator earnings depend on their assigned work volume and task difficulty.

Figure 2(b) shows that involved operators maintain an average capacity utilization of 60% throughout their operational activities. The model pushes several high performing individuals, such as operator 1, 11, 13 to 100% of their capacity. Operator 25 operates at only 11% capacity which indicates that some active operators serve to fulfill minor production needs rather than serving in core operational roles.

With target capacity level set to 10%, the model suggests involving all of the operators in the production while maintaining 10% of the overall capacity of the operators. Which makes the labor cost approximately 58,572 BDT and the income average and capacity average are BDT 1729 and 33% respectively.

Operations	SHOULDER JOIN	RIB MAKE	RIB SERVICING	NECK JOIN	BACK TAPE JOIN	NECK TOP STITCH	LABEL POSITION MARK	LABEL TOP STITCH	BODY FOLD & SLEEVE MATCHING	SLEEVE JOIN	ARMHOLE TOP STITCH	THREAD CUT & BODY FOLD	SIDE JOIN	THREAD CUT & TURN	SLEEVE HEM	THREAD CUT	BODY HEM	THREAD CUT	LABEL MAKE
Operators																			
operator 1	-	3800	-	-	-	-	-	-	-	-	-	-	-	-	10000	-	-	-	-
operator 2	-	-	-	-	-	-	-	10000	-	-	2644	-	-	-	-	-	-	-	-
operator 3	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	864	-	-
operator 4	-	-	-	-	-	-	-	-	2880	-	-	-	-	-	-	-	-	-	-
operator 5	-	-	-	509	-	-	-	-	-	762	-	-	-	-	-	-	-	-	-
operator 6	-	-	-	-	-	-	-	-	-	9238	-	-	-	-	-	-	-	-	-
operator 7	-	-	-	-	-	-	1520	-	1360	-	-	-	-	-	-	-	-	-	-
operator 8	-	-	-	-	-	-	2880	-	-	-	-	-	-	-	-	-	-	-	-
operator 9	-	500	-	-	-	-	-	-	-	-	-	-	-	-	-	-	4764	-	-
operator 10	-	-	-	-	-	-	-	-	2880	-	-	-	-	-	-	-	-	-	-
operator 11	-	-	-	9491	-	-	-	-	-	-	-	-	4350	-	-	-	-	-	-
operator 12	-	-	-	-	-	-	2880	-	-	-	-	-	-	-	-	-	-	-	-
operator 13	-	-	-	-	7552	-	-	-	-	-	-	-	502	-	-	-	-	-	-
operator 14	-	-	-	-	-	-	2220	-	-	-	-	660	-	-	-	-	-	-	-
operator 15	-	-	-	-	-	-	-	-	2880	-	-	-	-	-	-	-	-	-	-
operator 16	-	-	-	-	1440	-	-	-	-	-	-	-	-	-	-	-	-	-	-
operator 17	-	-	-	-	-	-	-	-	-	-	-	-	-	8320	-	-	-	-	-
operator 18	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	864	-	-
operator 19	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	3540	-	-	-
operator 20	1440	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
operator 21	-	-	-	-	-	-	500	-	-	-	-	-	-	1680	-	700	-	-	-
operator 22	-	-	-	-	-	-	-	-	-	-	7356	-	-	-	-	-	-	-	10000
operator 23	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	2880	-	-	-
operator 24	-	-	-	-	-	-	-	-	-	-	-	-	810	-	-	-	-	-	-
operator 25	7120	-	-	-	1008	-	-	-	-	-	-	-	3528	-	-	-	-	-	-
operator 26	-	4546	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
operator 27	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	2880	-
operator 28	-	1154	10000	-	-	10000	-	-	-	-	-	-	-	-	-	-	2144	-	-
operator 29	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	7120	-
operator 30	-	-	-	-	-	-	-	-	-	-	-	-	810	-	-	-	-	-	-
operator 31	1440	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
operator 32	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	864	-	-
operator 33	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	2880	-	-	-
operator 34	-	-	-	-	-	-	-	-	-	-	-	9340	-	-	-	-	-	-	-

Figure 3: Optimal task allocation (scenario 2)

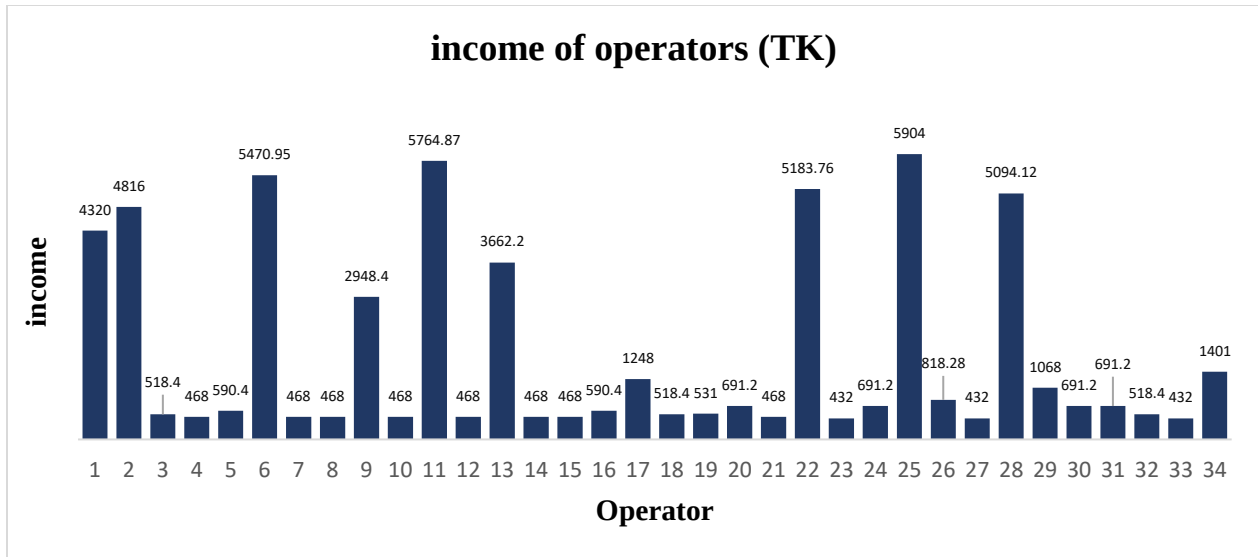


Figure 4(a): Income of operators (scenario 2)

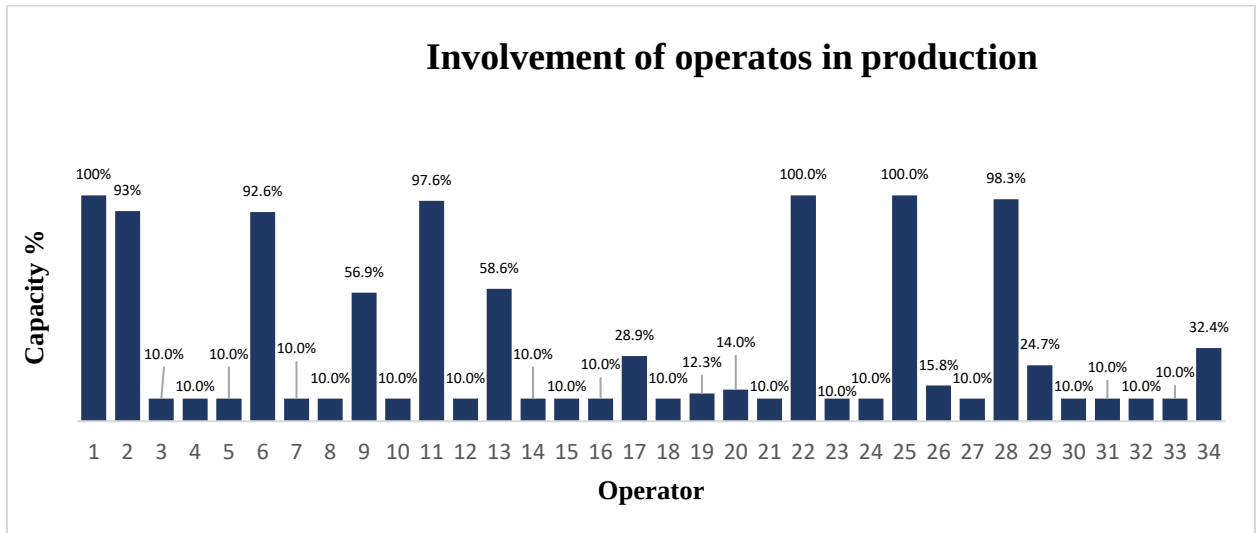


Figure 4(b): Capacity of operators (scenario 2)

Figure 4(a) represents the breakdown of earnings generated by each operator when the capacity is fixed at 10%. The average income per operator is 1729 BDT and the mandated 10% involvement ensures that even lowest utilized operators maintain a steady baseline income.

Figure 4(b) provides an understanding of how the fixed capacity constraint influences operator utilization. The data confirms that the 10% minimum threshold acts as a successful stabilizer, as a significant portion of the workforce operates precisely at the baseline. The homogeneity indicates the effectiveness of the task distribution that would normally have been performed by

fewer people. In addition, Figure 4(b) illustrates that there is still a section of employees operating at 100 percent capacity implying that there is no further division of the task without compromising its quality. The average capacity of 33 percent implies the importance placed on labor sustainability as opposed to intensity.

4.3 Discussion

The findings of the Binary Integer Linear Programming model reveal the ability of incorporation of skill-level heterogeneity and wage differentials into assignment of tasks in garment manufacturing. Using the model in a real life case of production in the Noman Group, the study finds the best allocation plans that will cut down the labor cost by a large margin over the current method of manual use in the factory, and at the same time meet all the operational requirements.

A major critique of the model is its direct comparison with the current practice of Noman Group in terms of manual allocation which costs a total labor cost of 68,832 BDT of the same production order. The scenario 1 of the BILP model (Cap = 0%) lowers this to about 58,131 BDT a cost saving of more than 10,700 BDT, which is almost 15.6 percent less.. Scenario 2 (Cap = 10%) will have a similar saving of about 10,260 BDT, which will decrease the cost to 58,572 BDT or about 15 percent of the current base. This empirical comparison directly proves that informal, experience-based allocation that is followed in Noman Group nowadays leaves a lot of potential cost-saving that remains untapped.

Regarding the workforce equity aspect, as the two situations are compared, there is an important managerial insight. Scenario 1 uses only 17 out of the 34 operators, focusing workloads on the most cost-efficient capable individuals and straining operators Scenario 2, by comparison, uses all 34 operators at only a marginal extra cost of 441 BDT in comparison to Scenario 1. This insignificant cost variance is evidence of near complete workforce engagement that fosters labor equality and sustainability of employment in the long-run.

In general, the paper has shown that a structured BILP model, which could be used as a convenient and feasible decision support aid to the garment SME planners. It offers production

managers a clear and repeatable model to base allocation decision-making that would meet cost limitations, workforce sustainability objectives, and actual factory environment.

CHAPTER 5

CONCLUSION & FUTURE WORK

5.1 Conclusion

This case study used a Binary Integer Linear Programming (BILP) to optimize task allocation, and minimize labor cost in garment manufacturing, based on the case of Noman Group in Bangladesh. Intlinprog solver of MATLAB was used to effectively allocate the production activities of 34 operators and 19 assembly parts with operational constraints being met. The results of the BILP model are used to bring out the strategic and operational choices in the garment manufacturing process. The model assists production managers to determine the best task allocation, which shows the individual operator potential and wage rate. The systematic method of assigning operators to assembly jobs allows garment, SME's to save on labor costs without any knowledge in programming or operational research. Also, flexibility of the model by the capacity parameter can be useful in contributing positively to equity in workforce by making all the operators involved in the production.

In order to simplify the model, certain simplifying assumptions have been taken into consideration, such as the model does not take into consideration uncertainty in cycle times or variations in demand and as such, this may have an impact on the reliability of the allocation decisions made in the real world production settings. The compatibility of skill of the workers was only represented by a numerical skill difficulty that was proportional to the scale without taking into account other aspects such as individual learning rates, operator fatigue or variation in output quality across different tasks.

5.2 Future Work

Future research recommendations involve further development of the model to include the aspect of uncertainty through stochastic or robust optimization methods. The framework would be more validated by testing it against other types of garment products and other factory settings or economic circumstances. Also, replacing or enriching the objective function with more extensive social and environmental measures, including worker wellbeing and overtime opportunity can increase the applicability of the model to actual world sustainability

evaluations. Finally converting the model into a convenient application interface, like an excel or web based application, can provide extra convenience to use and adoption by more production planners who do not have a technical background.

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